

Important Git commands	Description	Command
	Set a user name	<code>git config --global user.name "Mitesh Soni"</code>
	Set email address	<code>git config --global user.email "xxxxxxx@xxxxx.com"</code>
	Initialise a Git repository	<code>git init</code>
	Clone a repository	<code>git clone https://github.com/xxxxxxx/Angular9786-DevOps</code>
	Verify the status of files in the repository	<code>git status</code>
	Add files in the current directory for Commit operation	<code>git add .</code>
	Commit files	<code>git commit -m "First commit in the repository"</code>
	List your branches	<code>git branch</code>
	Switch branch	<code>git checkout</code>
	Create a new branch	<code>git branch new-branchname</code>
	Merge current branch with the target branch	<code>git merge target-branch</code>
	Show all commits	<code>git log</code>
	Save modified and staged changes	<code>git stash</code>
	Fetch and merge any commits from the remote branch	<code>git pull</code>

Jenkins

Jenkins is an open source tool that provides integration with existing tools used in application life cycle management to automate all activities, starting from build to test and deploy. Many beginners start their journey of automation using Jenkins. It is easy to learn and has a huge community for support.

Jenkins offers a simple way to create a pipeline that includes continuous integration and continuous delivery. Plugin based architecture helps to use existing tools in the pipeline; hence, too many changes in culture can be avoided in one go.



Figure 2: Jenkins Blue Ocean

Initial release	2011
Stable release	2.324
Written in	Java
Licence	MIT
Website	www.jenkins.io
GitHub repository	https://github.com/jenkinsci/jenkins
Features	- Jenkins installation is supported for Docker, FreeBSD, Gentoo, MacOS X, OpenBSD, OpenSUSE, Red Hat/Fedora/CentOS, Ubuntu/Debian, Windows, and Generic Java package (.war) - Multiple views are supported along with folders to manage jobs in an easier manner - New UI for Jenkins dashboard that is easy to use - New UI to create Pipeline as Code in an easier way - Pipeline as Code/multibranch pipeline to manage pipelines easily in repositories - Distributed architecture to manage Jenkins effectively - Role based access to users based on build jobs/folders/ pipelines/configurations/actions - Authentication - Integration with active directory, LDAP and with internal database
How is it useful in implementing DevOps practices?	Automation server helps to automate activities involved in application life cycle management
Can we integrate it with Pipeline as Code?	Yes; Pipeline as Code using Jenkinsfile
Is a commercial flavour available?	CloudBees Jenkins
Awards	- DevOps Dozen 2017: Best DevOps Open Source Project - DevOps Dozen 2016: Best DevOps Open Source Project - Bossie Awards 2014: The best open source application development tool - Geek Choice Award 2014 - SDTimes 100, 2014: DevOps & SCM - SDTimes 100, 2013: Development Tools & CI - SDTimes 100, 2012: Development Tools & CI - InfoWorld Bossies Award, 2011 - O'Reilly Open Source Award, 2011 - SDTimes 100, 2011: ALM & SCM - SDTimes 100, 2010: ALM Platforms - GlassFish Community Innovation Award 2008 - Duke's Choice Award
Third party integrations	- More than 1500 plugins to integrate Jenkins with many open source and commercial tools (https://plugins.jenkins.io/) - Platforms (iOS development, .NET, Android development, Ruby development) - User interface (User Interface, List view column plugins) - Administration (Agent controllers, Page decorators, Users and security, Cluster management, CLI extensions) - Source code management (SCM connections, SCM related) - Build management (Build triggers, Build wrappers, Build notifiers, Deployment plugins, Build parameters, Clean-up actions, Build tools, Build reports, Artifact uploaders)